Pseudo Complex-Valued Deformable ConvLSTM Neural Network With Mutual Attention Learning for Hyperspectral Image Classification

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***Abstract*— Convolutional long short-term memory (ConvLSTM) has received much attention for hyperspectral image (HSI) classification due to its ability of modeling long-range correlations, which, however, is vulnerable to too many parameters and insufficient training, limiting its classification accuracy, especially for small samples. Different from it, traditional hand-crafted methods extract the features with basic attributes of HSIs, which can provide the lack of details and interpretability of deep semantic features. However, existing methods fail to incorporate their complementarity for HSI classification. As such, a Pseudo complex-valued (CV) Deformable ConvLSTM Neural Network with mutual Attention learning (APDCLNN) is proposed, providing a new way to realize the collaborative learning of hand-crafted and deep features for HSI classification. First, a 2-D pseudo CV deformable ConvLSTM (PDConvLSTM2D) cell is designed using deformable convolution and complex operations, with which a spatial–spectral PDConvLSTM2D neural network (SSPDCL2DNN) is built to extract scale- and spectral-enhanced deep spatial–spectral features. Then, 3-D Gabor filter is used to extract hand-crafted features, and a mutual attention-based multimodality feature learning and fusion (MAMLF) module is designed to integrate them into deep features for training and optimization of SSPDCL2DNN. Finally, an attention loss subnetwork is designed to refine the classification results. As we know, this is the first attempt to apply the idea of mutual attention learning to fuse hand-crafted and deep features for HSI classification. Extensive experiments on three widely used HSI datasets show the advantages of our model over other deep methods in terms of both quantitative and visual quality.**

***Index Terms*— Complex-valued (CV) representation, convolutional long short-term memory (ConvLSTM), feature fusion, hyperspectral image (HSI) classification, mutual attention learning.**

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I. Introduction

HYPERSPECTRAL images (HSIs) present a 3-D data cube with rich spatial and spectral information of land covers, which have been applied in various applications, i.e., geological exploration, precision agriculture, and ecological science [1].

Joint learning of spatial–spectral information has a vital effect on HSI classification. Different from the graph embedding algorithms [2], [3], the simplest way is to introduce a set of predesigned 3-D filters and interact with HSI data to extract the spatial–spectral features. Shen and Jia [4] used the 3-D Gabor filter to learn the features about local signal changes in spatial, spectral, and spatial–spectral domains for HSI classification. A pixel classification method was designed by combining structured sparse logistic regression and discrete wavelet transform [5]. A multiple feature fusion framework was designed by combining 3-D Gabor surface feature, morphological profile [6], and local binary pattern (LBP) [7] for HSI classification [8]. Other 3-D filters, i.e., scattering wavelet transform [9], scale-invariant feature transform [10], and multidimensional LBP [11], were also developed. These studies have shown that hand-crafted features can provide low-level and interpretable spatial–spectral features with different frequencies, orientations, attributes, and modalities for HSI classification.

Recent years have witnessed the success and promising results of deep learning in the fields of computer vision and remote sensing, and various deep neural networks (DNNs) have been emerged [12], such as convolutional neural networks (CNNs), capsule networks (CapsNets), and graph neural networks [13]. Different from them, recurrent neural networks (RNNs) [14] have shown their ability in handling sequence data, especially long short-term memory (LSTM) [15] and convolutional LSTM [16] that is renamed as ConvLSTM2D in [17]. Then, some deep frameworks using them alone or in combination with CNNs have been proposed to learn the correlations of time series, such as the CNN+LSTM structures for Siamese tracking [18] and sentiment detection [19], the CNN+ConvLSTM2D models for speech emotion recognition [20], image segmentation [21], and trajectory prediction [22]. By considering that special

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imaging mechanism makes HSI data contain strong spectral correlation, some works have applied the LSTM or ConvLSTM2D cell as a basic unit to capture it for HSI classification, such as bidirectional-ConvLSTM2D [23], 3-D convolution-recurrent network [24], cascaded RNN [25], spatial–spectral ConvLSTM2D neural network (SSCL2DNN) [17], CapsNet+ConvLSTM2D [26], and tensor-attention-driven ConvLSTM neural network (TACLNN) [27]. Furthermore, a new spatial–spectral ConvLSTM3D was extended from ConvLSTM2D to learn the structure information of HSIs [17], and some ConvLSTM3D-based models have been proposed, such as SSCL3DNN [17], ConvLSTM3D+attention mechanism (dual-channel A^3CLNN) [28], and ConvLSTM3D+3-D CNN [spectral-spatial 3-D ConvLSTM-CNN based residual network (SSCRN)] [29]. To enhance the scale-aware ability of convolution operation, a deformable convolutional layer was designed to enhance CNNs [30], with which a deformable CNN was proposed for HSI classification [31]. Based on the above analysis, these DNNs can automatically extract high-level, internal, and abstract semantic features for HSI classification.

The above works have shown that both these modality features (e.g., hand-crafted and deep features) can contribute to HSI classification. However, DNNs usually required many labeled data to optimize their underlying parameters and learn the representations with multilevel abstractions, leading to overfitting and limited performance especially under the case of small samples [32]. Some works have analyzed the characteristics of these two features and found that hand-crafted features are helpful to supervising the training process and reducing the impacts of overfitting on DNNs [33]–[35]. Some researches have tried to use hand-crafted filters to improve the inputs [36] or weights [37] of DNNs; however, the effects of integrating hand-crafted and deep features on classification performance have not been fully explored for HSIs. In particular, for HSI classification, the available labeled samples are usually limited and obtained at expensive labeling costs. Therefore, it is of great significance to investigate their complementarity and integrate hand-crafted features into DNNs to improve their classification performance for HSIs.

Nowadays, most of the DNNs are built in the real number domain; however, it has been validated that the complex number operation can provide advantageous representation ability than the real number one, and some researchers have analyzed the mathematical motivations of complex number operation on DNNs [38]–[43]. There are two different setups for complex-valued (CV) DNNs. The first setup is the fully CV-DNNs, where both inputs and DNNs are CV format, which have been used for polarimetric synthetic aperture radar (PolSAR) image classification [44], [45] and SAR target classification [46]. Different from the above works, there is a new setup, where using the advantages of CV representation in improving the feature learning ability, the real-valued (RV) data are converted into their CV formats using some methods for CV-DNNs. We call it pseudo CV-DNNs. The complex number operations were applied to improve DNNs (e.g., LSTM, CNNs), where the first half and second half of the RV vectorized image [47], the feature maps along their channel dimensions [48], [49] or different RV filters [50] are taken as the real and imaginary parts, respectively. Particularly, Peker *et al.* [51] were the first to use deep features from CNN to build a CV attribute set where every two features are randomly selected to form a complex number and used a CV wavelet neural network [52] to predict its results for HSI classification. The above works illustrate that CV-DNNs can achieve richer representational ability, faster optimization efficiency, and competitive performance over their RV counterparts.

Since attention mechanism was first introduced into deep learning, many attention-driven deep models have been built for various applications, where a hard part selection subnetwork or a soft mask branch was its most common use, such as spatial–channel [53], spatial–temporal [54], spatial–spectral and multiscale [28], and tensor [27] attention modules. Moreover, to learn and fuse the complementary correspondences of multimodality data or features, the idea of mutual attention learning has been further proposed. The cross-modal attention modules were designed to capture the interdependencies for object detection in red, green, and blue (RGB)-depth images [55] or image-text retrieval [56], while a co-attention module was applied to learn correlations of different features for salient detection of image pair [57]. Han *et al.* [58] developed a mutual attention module to adaptively use the special information of knowledge graph and text to mutually guide their representation learning. In addition, the mutual attention mechanism was also applied to adaptively exchange the information and capture the mutual impact of different features from different DNNs [59] or layers of CNNs [60]. Specifically, as for HSI classification task, the complementarity between multisource data or multifeature representations is beneficial to improve the classification accuracy, which can also be viewed as the multimodality data or features. However, as far as we know, the idea of mutual attention learning has not been considered to integrate the multimodality features for HSI classification.

Witnessing the independent success of hand-crafted and deep features for HSI classification [4]–[11], [23]–[28], [31], and based on our previous work in [17], a Pseudo CV Deformable ConvLSTM Neural Network with mutual Attention learning (APDCLNN) is proposed. First, using deformable convolution and complex operations, a pseudo CV deformable ConvLSTM2D (PDConvLSTM2D) cell is developed, with which a spatial–spectral PDConvLSTM2D neural network (SSPDCL2DNN) is built to extract scale- and spectral-enhanced deep features from HSIs. Then, to integrate hand-crafted features into deep features to assist the training of the whole model, a mutual-attention-based multimodality feature learning and fusion (MAMLF) module with a 3-D Gabor feature extraction block (GFEB) and a mutual attention fusion block (MAFB) is further designed, where an attention loss subnetwork is built to adaptively refine the classification results. In summary, the main contributions of this work involve the following three aspects.

1) To enhance the abilities of modeling the scale information and long-range spectral correlation, a new PDConvLSTM2D cell is developed by jointly using the advantages of deformable convolution and complex number operations, with which an

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SSPDCL2DNN is designed to extract better deep features from HSIs.

2) An MAMLF module is designed to explore and fuse the fine-grained complementarity of multimodality spatial–spectral features, from which an attention loss subnetwork is built. It is used in conjunction with the classification loss subnetwork for HSI classification.

3) By simultaneously considering the contributions of hand-crafted and deep features, an APDCLNN is further proposed, where hand-crafted features are applied to supervise its training process and reduce the impact of overfitting from limited samples, thus adaptively exchanging the information and mutual effects of different features for better classification results of HSIs.

The rest of this article is organized as follows. In Section II, we briefly introduce some related works. Section III elaborates the framework of APDCLNN in detail. Section IV introduces the hyperparameter settings, experimental results, and analysis, followed by summarizations in Section V.

II. Related Work

A. ConvLSTM

ConvLSTM2D was built by extending the way of input-to-state and state-to-state transitions in LSTM to 2-D convolution for spatio-temporal learning [16], whose information protection and control are executed by gate structures (i.e., input gate i_t, forget gate f_t, output gate o_t). Its formulas are written as

$$\begin{aligned} i_t &= \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + W_{ci} \circ C_{t-1} + b_i) \\ f_t &= \sigma\left(W_{xf} * X_t + W_{hf} * H_{t-1} + W_{cf} \circ C_{t-1} + b_f\right) \\ \tilde{C}_t &= \tanh(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c) \\ C_t &= f_t \circ C_{t-1} + i_t \circ \tilde{C}_t \\ o_t &= \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + W_{co} \circ C_t + b_o) \\ H_t &= o_t \circ \tanh(C_t) \end{aligned} \tag{1}$$

where X_t means the input of the current cell. C_{t-1} and H_{t-1} are, respectively, the state and output of the last cell. Here, $W_{\cdot i}$, b_i, $W_{\cdot f}$, b_f, and $W_{\cdot o}$, b_o are, respectively, the weights and biases of three gates, in which $\cdot$ stands for x, h, or c. $*$ and $\circ$ refer to the convolution operation and the Hadamard product, respectively. $\sigma(\cdot)$ and $\tanh(\cdot)$ are the activation functions.

With the ability of modeling long-term dependencies in the time, spectral, or other domains, many ConvLSTM2D-based models have been proposed using it alone or in combination with CNNs for various applications, such as Siamese tracking [18], text sentiment detection [19], speech emotion recognition [20], image segmentation [21], trajectory prediction [22], and HSI classification [17], [23]–[29].

B. Deformable CNN

The scale-fixed convolution operation is usually used in CNNs. To enhance their transformation modeling ability, a deformable convolutional layer was designed using additional offsets automatically learned from specific tasks [30], which has been used for different applications, such as HSI classification [31], object tracking [61], and person reidentification [62]. Its calculation formulas are expressed as

$$\begin{aligned} \Delta p_n &= off(x) \\ y(p_0) &= \sum_{p_n} w(p_n) x(p_0 + p_n + \Delta p_n) \end{aligned} \tag{2}$$

where p_0 is each location in the output y calculated by the input x and weight w, $off(\cdot)$ is the function to learn the offset Δp_n, p_n contains all the locations in the receptive field of w, and n denotes the number of these locations.

From (2), the deformable convolution operation is calculated as an irregular location with offset $p_n + \Delta p_n$, which is the main reason that it can adaptively capture the multiscale contextual features from input, and defined as $y = w \tilde{*} x$ for convenience with $\tilde{*}$ being the deformable convolution operation.

C. CV-CNN

By considering that complex number has easier optimization, better generalization characteristics, and faster learning than real number, various CV-DNNs were designed for CV or RV data analysis and process [44]–[51], obtaining better performance than their RV counterparts for different applications.

For pseudo CV-CNNs, given a RV 2-D convolutional layer (formulated as $Y = W * X$) with a $k \times k \times C \times S$ weight, the first $(C/2)$ features are used as the real part while the remaining are used as the imaginary part, where k, C, and S are, respectively, the kernel size, the numbers of input, and output channels. Therefore, the CV convolutional layer is defined as

$$\begin{aligned} Y^R + jY^I = &\left(W^R * X^R - W^I * X^I\right) \\ &+ j\left(W^I * X^R + W^R * X^I\right) \end{aligned} \tag{3}$$

where $W = W^R + jW^I$ is the CV weight. $X = X^R + jX^I$ and $Y = Y^R + jY^I$ are the CV input and output, respectively, R denotes the real part, and I means the imaginary part.

However, in [47]–[51], RV-DNNs were directly replaced with their CV counterparts, without corresponding designs according to the structures of inputs or RV-DNNs. In addition, the input data or features were divided into two parts with some methods to construct the complex number, which may destroy their inherent structures to a certain extent.

D. Attention Mechanism

Since attention mechanism was first applied to improve the deep models, many attention-driven models have been built for various applications [27], [28], [53], [54].

With the developments of internet and computer technology, multimodality data (i.e., image, text, audio) or features from different learning methods are widely available, which is also one of the methods to obtain better performance, and how to learn effective representations from multimodality data or features has become a thorny challenge. As such, some efforts have been made with mutual attention learning to model the complementarity of multimodality data or features. For co-saliency detection, the cross-modal attention structure was used to work as the guidance of attention to one modality on the another modality to acquire strong representations from

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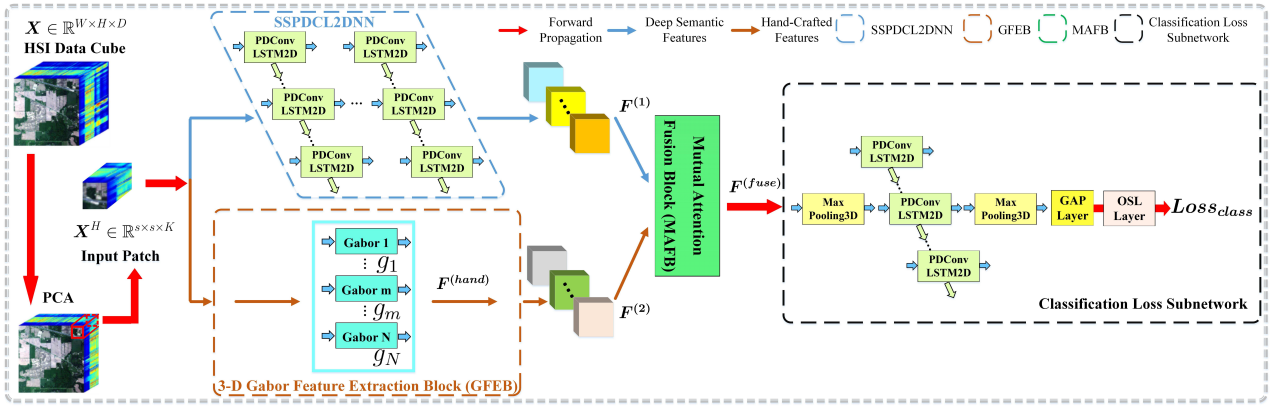

Fig. 1. Flowchart of the proposed APDCLNN model.

RGB-D or pairwise images [55], [57]. For image-text retrieval, it was applied to perform attentions on both the modalities to drive the deep models to learn visual–textual interdependencies [56]. [58]–[60] developed the mutual attention module to explore the special information of multimodality features from different methods to adaptively exchange their respective information and mutually guide their representation learning. Therefore, the idea of mutual attention learning has been a useful method to produce multiply attention maps to interactively reweight the multimodality data or features for effectively incorporating their complementary correspondences.

For HSI classification, the joint learning of multisource data [i.e., HSI and light detection and ranging (LiDAR)] or multifeature representations (i.e., hand-crafted features and deep semantic features) can improve the classification accuracy of HSIs, which can also be seen as a learning task of multimodality data or features. Therefore, mutual attention learning can also be useful to capture the fine-grained complementarity of multimodality data or features for HSI classification.

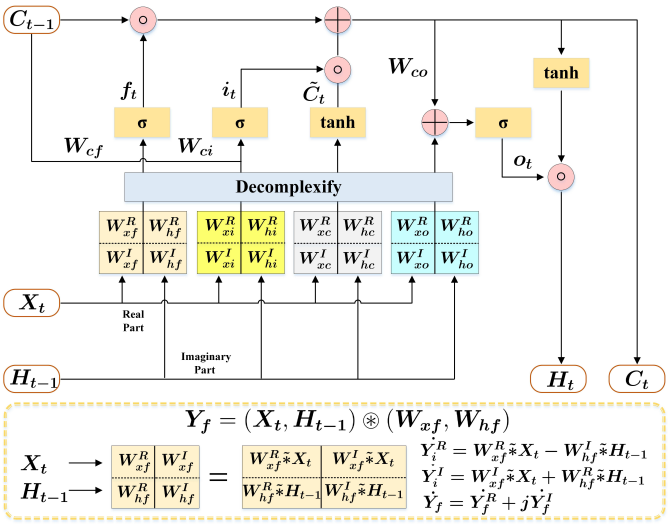

Fig. 2. Structure of the developed PDConvLSTM2D cell. "Decomplexify" refers to convert the data into a real number.

III. APDCLNN

A. Architecture Overview

Fig. 1 shows the overview of our APDCLNN. To enhance the feature extraction ability and maintain the inner structure of input data in ConvLSTM2D, a PDConvLSTM2D cell is developed in Section III-B. In Section III-C, an MAMLF module is presented to explore and fuse hand-crafted and deep features of HSIs. Finally, Section III-D outlines the framework of our APDCLNN in detail.

To the best of our knowledge, this is the first application of mutual attention learning to learn the complementary information and mutual effects between multimodality features (i.e., hand-crafted features and deep features) and supervise the training process and reduce the impact of overfitting from limited samples of our APDCLNN for HSI classification.

B. PDConvLSTM2D Cell

Inspired by Sections II-B and II-C, to improve the abilities of scale feature extraction and correlation learning of ConvLSTM2D for HSI classification, by combining the ideas of deformable convolution and pseudo CV-CNNs, a PDConvLSTM2D cell is designed, as shown in Fig. 2.

First, to adapt to the multiscale characteristics of different land covers in HSIs and improve the ability to model the geometric transformations of the ConvLSTM2D-based models, a 2-D deformable ConvLSTM (DConvLSTM2D) cell is built, where the convolutional layers in each ConvLSTM2D cell are improved to the deformable ones. Particularly, similar to [30], [31], [61], a 3×3 convolutional layer is set as the function $off(\cdot)$ to calculate the offset Δp_n in (2).

To further improve the feature extraction capacity for long-term spectral correlation of DConvLSTM2D, inspired by the advantages of CV operations [38]–[50] and the first work of complex number for HSI classification [51], a pseudo CV DConvLSTM2D (PDConvLSTM2D) cell is developed for HSI classification. Particularly, unlike using the CV operations to replace the RV counterparts directly for RV data analysis and process in the pseudo CV-DNNs [47]–[51], each gate of the PDConvLSTM2D cell is improved by the complex number operation and representation, where the input information X_t in (1) is regarded as the real part, while the recurrent

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information H_{t-1} is used as the imaginary part. Take the input gate as an example. Based on (2) and (3), the CV output $\dot{Y}_i$ of the first two items in (1) becomes

$$\begin{aligned} \dot{Y}_i^R &= W_{xi}^R \tilde{*} X_t - W_{hi}^I \tilde{*} H_{t-1} \\ \dot{Y}_i^I &= W_{xi}^I \tilde{*} X_t + W_{hi}^R \tilde{*} H_{t-1} \\ \dot{Y}_i &= \dot{Y}_i^R + j\dot{Y}_i^I \end{aligned} \tag{4}$$

where $\dot{Y}_i^R$ and $\dot{Y}_i^I$ are the real and imaginary parts corresponding to $\dot{Y}_i$, which is then converted into the real value as the final output (e.g., Y_i) with the same operation as [48], [49], and a 1×1 convolutional layer is executed for dimension matching. The above calculation process is defined as

$$Y_i = (X_t, H_{t-1}) \circledast (W_{xi}, W_{hi}) \tag{5}$$

where $\circledast$ is the CV deformable convolution operation.

Based on the above analysis, the calculation formulas of the whole PDConvLSTM2D cell can be updated to

$$\begin{aligned} i_t &= \sigma((X_t, H_{t-1}) \circledast (W_{xi}, W_{hi}) + W_{ci} \circ C_{t-1} + b_i) \\ f_t &= \sigma((X_t, H_{t-1}) \circledast (W_{xf}, W_{hf}) + W_{cf} \circ C_{t-1} + b_f) \\ \tilde{C}_t &= \tanh((X_t, H_{t-1}) \circledast (W_{xc}, W_{hc})) + b_c) \\ C_t &= f_t \circ C_{t-1} + i_t \circ \tilde{C}_t \\ o_t &= \sigma(X_t, H_{t-1}) \circledast (W_{xo}, W_{ho}) + W_{co} \circ C_t + b_o) \\ H_t &= o_t \circ \tanh(C_t). \end{aligned} \tag{6}$$

By taking it as a basic unit, a PDConvLSTM2D layer is further obtained. Different from the traditional pseudo CV-DNNs [47]–[51], in our PDConvLSTM2D layer, the real and imaginary parts come from the current and recurrent information of each basic unit, and all the weights in (4) are independent. In addition, the inner structure of data and the characteristics of the original ConvLSTM2D layer can be effectively maintained. Therefore, the feature extraction abilities for the scale information and the spectral correlation in our designed PDConvLSTM2D layer can be improved by exploiting the advantages of deformable convolution and complex number operation and representation without damaging the inherent structure of the input data or network.

In particular, an SSPDCL2DNN model is obtained by replacing each ConvLSTM2D layer in SSCL2DNN [17] with our PDConvLSTM2D layer for deep feature extraction of HSIs. For convenience, the output of the lth PDConvLSTM2D layer is defined as $F^{(1)} \in \mathbb{R}^{K \times w \times h \times C}$, where K, w, and h are the number of spectral bands, width, and height, respectively.

C. MAMLF Module

The small number of labeled samples from costly sample labeling fails to complete the full training and optimization of SSPDCL2DNN, limiting its performance in HSI data analysis. Some works have verified that hand-crafted features can contribute to HSI classification [4]–[11], and the incorporation of hand-crafted and deep features can be beneficial to assist the training and adjust the optimization direction of DNNs to some degree [32]–[35]. It may be a technology to improve the classification accuracy of SSPDCL2DNN for HSIs, especially in the situation of limited training samples. Therefore, to effectively supervise the training process and reduce the adverse impacts of overfitting on SSPDCL2DNN, inspired by the idea of manual attention learning [55]–[60], a ConvLSTM3D-based MAMLF module with the GFEB and MAFB submodules is built to integrate hand-crafted features extracted using the 3-D Gabor filter into deep features from SSPDCL2DNN to assist the training process of the whole model, improving its classification performance for HSIs.

Let the original HSI data be denoted as $X \in \mathbb{R}^{W \times H \times D}$, where W, H, and D are the width, height, and the number of spectral bands, respectively. For each pixel x, the neighboring region with a size of $s \times s$ is extracted as its spatial information. Similar to [17], principal component analysis (PCA) is applied to reduce data redundancy and computational complexity, and the top K components are maintained as the spectral information of x, forming a 3-D volume (i.e., $X^H \in \mathbb{R}^{s \times s \times K}$).

First, due to the fact that the 3-D Gabor filter can directly process the HSI data to characterize the low-level spatial–spectral features about local signal changes in spatial, spectral, and joint spatial–spectral correlations [4], a 3-D Gabor-based GFEB submodule is built, as shown in Fig. 1 with brown dotted line. Specifically, X^H is convolved with a set of 3-D Gabor wavelets with different frequencies f and orientations θ and ψ to extract hand-crafted spatial–spectral features as

$$F^{(\text{hand})} = \left[g_1\left(X^H\right), \ldots, g_m\left(X^H\right), \ldots, g_N\left(X^H\right)\right] \tag{7}$$

where $F^{(\text{hand})} \in \mathbb{R}^{K \times w \times h \times N}$ and $g_m(\cdot)$ for $m = 1, 2, \ldots, N$ is the 3-D Gabor wavelet. For the convenience of subsequent analysis, the extracted hand-crafted feature is defined as $F^{(2)}$. Based on the above analysis, the multimodality spatial–spectral features (i.e., deep feature $F^{(1)}$, hand-crafted feature $F^{(2)}$) can be obtained from HSI data.

To fuse them, an MAFB submodule is further designed, as shown in Fig. 3 with green dotted line. Because their dimensions may be inconsistent, a rectified linear unit (ReLU) function, a $3 \times 3 \times 3$ ConvLSTM3D layer [17], and a softmax function are executed to generate their own identity maps and attention maps as

$$\begin{aligned} F^{(1)'} &= f_{CL3D3}\left(\text{ReLU}\left(F^{(1)}\right)\right) \\ F^{(2)'} &= f_{CL3D3}\left(\text{ReLU}\left(F^{(2)}\right)\right) \\ \alpha^{(1)} &= \text{softmax}\left(F^{(1)'}\right) \\ \alpha^{(2)} &= \text{softmax}\left(F^{(2)'}\right) \end{aligned} \tag{8}$$

where $F^{(1)'}$ and $F^{(2)'} \in \mathbb{R}^{K \times w \times h \times C'}$ are the identity maps of these two modalities, $\alpha^{(1)}$ and $\alpha^{(2)}$ are their attention maps, and C' is the channel number after feature matching. Then, the identity map of one modality is interactively reweighted by the attention map of the paired modality. The fused spatial–spectral feature $F^{(1)(2)}$ is written as

$$\begin{aligned} \hat{F^{(1)}} &= F^{(1)'} + \alpha^{(2)} \otimes F^{(1)'} \\ \hat{F^{(2)}} &= F^{(2)'} + \alpha^{(1)} \otimes F^{(2)'} \\ F^{(1)(2)} &= \hat{F^{(1)}} + \hat{F^{(2)}} \end{aligned} \tag{9}$$

where $\otimes$ is the element-wise multiplication operation.

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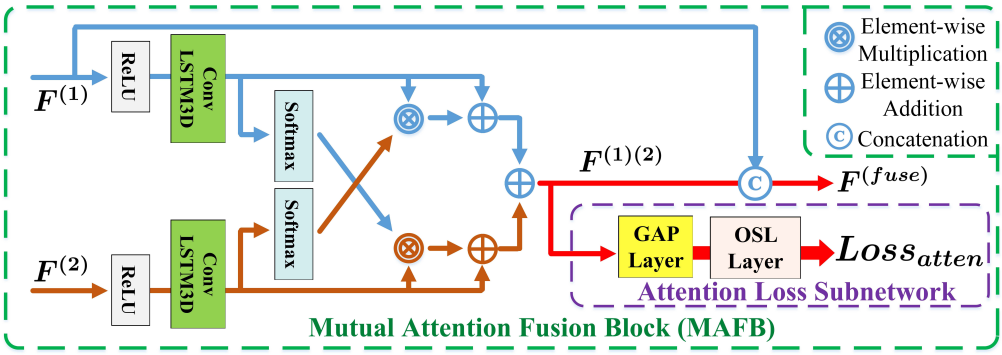

Fig. 3. Structure of the proposed MAFB submodule.

To pursue more accurate predictions, inspired by [57], skip connection is applied to provide more abundant information for HSI classification. In addition, similar to [63], no layer is used before merging these two features, except element-wise summation or concatenation operation, which can preserve the relationships between different features. Therefore, the final output of MAMLF is improved as $F^{(fuse)} = [F^{(1)}, F^{(1)(2)}]$, where $[\cdot, \cdot]$ is the concatenation operation. Based on the above design, MAMLF can encourage the whole model to discover complementary evidences besides the most dominant ones by projecting these two features with different modalities from their own feature spaces into one single common space equally and effectively integrate the traditional hand-crafted features into deep semantic features by dynamically adjusting the relative weight of each modality, thus generating more expressive spatial–spectral features for HSI classification.

Finally, aiming at the overfitting problem in small-sample classification, Li *et al.* [64] proposed an orthogonal softmax layer (OSL), a classification layer with orthogonal weight during the training and testing processes. Hence, a novel attention loss subnetwork with a global average pooling (GAP) layer and an OSL layer is developed from MAMLF, which is marked with purple dotted line in Fig. 3 to mitigate the overfitting and refine the classification results adaptively, where $F^{(1)(2)}$ is reused as the input to build an attention loss function (namely $Loss_{atten}$) with the cross entropy. The forward propagation of this part is written as

$$\begin{aligned} F^{(atten)} &= f_{OSL}\left(f_{GAP}\left(F^{(1)(2)}\right)\right) \\ Loss_{atten} &= -\sum\left(Y \cdot \log\left(F^{(atten)}\right)\right) \end{aligned} \tag{10}$$

where $f_{OSL}(\cdot)$ and $f_{GAP}(\cdot)$ mean the OSL and GAP layers, respectively. $F^{(atten)}$ and Y are the output of our attention loss subnetwork and the ground-truth label, respectively.

D. APDCLNN Model

To reduce the adverse impacts of overfitting for improving the classification accuracy of HSIs, an APDCLNN model is built by combining SSPDCL2DNN and MAMLF to explore and fuse the complementary information between hand-crafted and deep features, as shown in Fig. 1.

First, similar to the preprocessing method in Section III-C, the 3-D volume $X^H \in \mathbb{R}^{s \times s \times K}$ of pixel x is extracted from the original HSI data X, which is the input of SSPDCL2DNN and the MAMLF module. Then, from Section III-B, the backbone of SSPDCL2DNN is obtained by alternate connection of some PDConvLSTM2D and max-pooling layers, in which τ_{2D} 2-D components are extracted by decomposing X^H and converted into a spectral sequence, i.e., $\{X^H_1, \ldots, X^H_{t_2}, \ldots, X^H_{\tau_{2D}}\}$. $t_2 \in \{1, 2, \ldots, \tau_{2D}\}$. τ_{2D} equals K, which is the dimension time_step in the PDConvLSTM2D layer. This sequence is fed into SSPDCL2DNN to extract deep spatial–spectral features, where the output $F^{(l)}$ of the lth PDConvLSTM2D layer is the extracted deep semantic feature. In our experiments, l is 1. Then, the 3-D Gabor filter is applied to convolve with X^H to extract low-level spatial–spectral features about local signal changes in the spatial, spectral, and joint spatial–spectral correlations, where the output $F^{(2)}$ of the GFEB submodule is the extracted hand-crafted feature. After obtaining these two features (i.e., $F^{(1)}$, $F^{(2)}$), their fusion representation $F^{(\text{fuse})}$ is generated by the MAFB submodule for collaborative learning and fusion of hand-crafted features and deep features for HSI classification.

After that, at the top of APDCLNN, a classification loss subnetwork is built in Fig. 1 with black dotted line, where $F^{(fuse)}$ is its input, and then two max-pooling layers and a PDConvLSTM2D layer are executed alternately to learn better representation. Similar to (10), to alleviate overfitting, a GAP layer and an OSL layer are further applied to replace fully connected (FC) and softmax layers to predict the conditional probability distribution of each class, where the cross entropy is used again as the classification loss function, marking as $Loss_{class}$.

Finally, by combining the above two loss functions (i.e., $Loss_{atten}$, $Loss_{class}$), a multitask loss function is built as

$$Loss_{mtask} = Loss_{class} + \lambda Loss_{atten} \tag{11}$$

where λ is the weighting factor to control these two losses.

To further reduce the occurrence of overfitting, $l2$ regularization and dropout operations are also considered as two additional regularization tricks. In addition, APDCLNN is trained in an end-to-end manner, and multitask loss function $Loss_{mtask}$ is optimized using the adaptive momentum algorithm with a learning rate lr to yield the final classification results. The training process is summarized in Algorithm 1.

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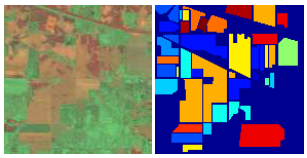
Indian Pines



(a)

NO.	Color	Class	Training	Total
1		Alfalfa	10	41
2		Corn-notill	10	1561
3		Corn-mintill	10	903
4		Corn	10	251
5		Grass-pasture	10	521
6		Grass-trees	10	793
7		Grass-pasture-mowed	10	21
8		Hay-windrowed	10	516
9		Oats	10	12
10		Soybean-notill	10	1059
11		Soybean-mintill	10	2691
12		Soybean-clean	10	642
13		Wheat	10	216
14		Woods	10	1382
15		Buildings-Grass-Trees-Drives	10	415
16		Stone-Steel-Towers	10	92
Total			160	11116

(d)

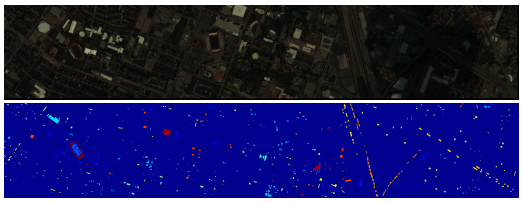
University of Pavia

(b)

NO.	Color	Class	Training	Total
1		Asphalt	10	6687
2		Meadows	10	18825
3		Gravel	10	2110
4		Trees	10	3085
5		Painted metal sheets	10	1348
6		Bare Soil	10	5069
7		Bitumen	10	1333
8		Self-Blocking Bricks	10	3709
9		Shadows	10	946
Total			90	43112

(e)

Houston Data



(c)

NO.	Color	Class	Training	Total
1		Health grass	10	1439
2		Stressed grass	10	1434
3		Synthetic grass	10	879
4		Tress	10	1422
5		Soil	10	1418
6		Water	10	497
7		Residential	10	1454
8		Commercial	10	1425
9		Road	10	1435
10		Highway	10	1408
11		Railway	10	1406
12		Parking lot 1	10	1415
13		Parking lot 2	10	643
14		Tennis court	10	599
15		Running track	10	837
Total			150	17711

(f)

Fig. 4. (Left) False-color maps and (Right) ground-truth maps of three HSI datasets. (a) Indian Pines (bands 20, 40, and 60). (b) University of Pavia (bands 47, 27, and 13). (c) Houston Data (bands 57, 27, and 17). (d)–(f) Number of the training samples.

Algorithm 1 Training APDCLNN for HSI Classification

Input: HSI data X; Ground truth Y
Output: Classification map Ω

Prepare the 3-D volume X^H of the pixel x
Parameter setting and weight initialization
Extract the traditional hand-crafted feature $F^{(2)}$
while step $\leq N_{steps}$ **do**
  Train the proposed APDCLNN model by optimizing the multitask loss function $Loss_{mtask}$
end while
return Classification map Ω

IV. Experimental Results

To evaluate the performance of our APDCLNN, several related spatial–spectral classification algorithms are adopted for comparison, i.e., SSCL2DNN and SSCL3DNN [17], 3DG-CNN [37], composite kernel support vector machine (SVM-CK) [65], principal-based convolutional neural network (PCNN) [66], and 3-D octave convolution with the spatial-spectral attention network (3DOC-SSAN) [67]. In addition, by removing the MAMLF module, APDCLNN is degenerated to an SSPDCL2DNN model, which will be changed into an spatial-spectral DConvLSTM2D neural network (SSDCL2DNN) model that is constructed by the ConvLSTM2D layer improved by deformable convolution [30]. Overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) are used as the quantitative metrics for assessment. By considering the deviations caused by random selection of training samples, the average value of each metric after ten Monte Carlo runs is reported. The proposed model is built with Tensorflow platform (i.e., Tensorflow-GPU 1.12.0, python 3.5.4) and trained on a desktop with an Intel Core i7-8700 processor and a Nvidia GeForce GTX 1080ti GPU.

A. Hyperspectral Data

Our APDCLNN is evaluated on three public HSI datasets, such as Indian Pines, University of Pavia, and 2013 Houston Data, whose detailed information are reported in Fig. 4.

1) Indian Pines: It was acquired in 1992 by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in Northwestern Indiana, USA, which contains 145×145 pixels with a spatial resolution of 20 meters per pixel (mpp) and 16 classes. Regardless of some unusable bands, 200 bands in the wavelength range from 0.4 to 2.5 μm are maintained.

2) University of Pavia: It was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over University of Pavia, Northern Italy, consisting of 610×340 pixels with a spatial resolution of 1.3 mpp and nine land-cover classes. In all, 103 spectral bands in the spectral range from 0.43 to 0.86 μm are retained by removing the noise-corrupted bands.

3) Houston Data: It was obtained in 2012 by the Compact Airborne Spectrographic Imager (CASI) sensor over University of Houston campus and its surrounding area, which was the official dataset in the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion contest. There are 349×1905 pixels with a spatial resolution of 2.5 m, 144 spectral bands in the wavelength range from 0.38 to 1.05 μm, and 15 labeled classes. This dataset is available online from the IEEE GRSS Data and Algorithm Standard Evaluation Website: `http://dase.grss-ieee.org/`.

B. Experimental Settings

All the parameters of the comparison methods are tuned according to [17], [37], [65]–[67]. However, due to memory limitation and similar to other methods, the PCA algorithm

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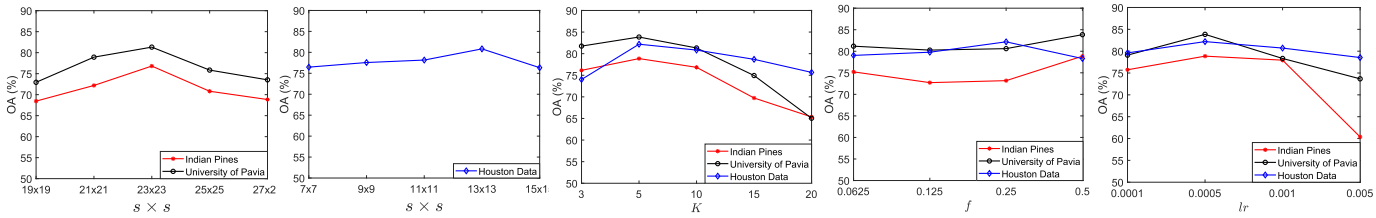

Fig. 5. OA (%) achieved by our APDCLNN model with different parameters for the three HSI datasets. (a)–(b) Window size $s \times s$. (c) Number K of the principal components. (d) Frequency f of the 3-D Gabor filter. (e) Learning rate lr.

is also applied to 3DOC-SSAN [67]. In addition, ten samples from each class in the three HSI datasets are randomly extracted for training, while the remaining samples for testing.

For our APDCLNN, there are nine key parameters to be determined, i.e., the local window size ($s \times s$), the number (K) of the principal components, the number (N) and frequency (f) of the 3-D Gabor filters in the GFEB submodule, the kernel size ($k \times k$) and the number (M) of the feature maps in each PDConvLSTM2D layer, the dropout value, the weight λ, and the learning rate lr. Initially, K is fixed to 10, N to 13, and f to 0.25. M is set to $\{32, 64\}$ for two PDConvLSTM2D layers. The value of λ is fixed to 0.5, the dropout value to 0.5, and lr to 0.0005. After that, s is selected from $\{19, 21, 23, 25, 27\}$ for the Indian Pines and University of Pavia datasets, while from $\{7, 9, 11, 13, 15\}$ for the Houston dataset. k is from $\{3, 4, 5\}$. The experiments for analyzing the influence of different sizes of local window on classification performance are conducted in Fig. 5(a) and (b). We can observe that the optimal value of $s \times s$ is 23×23, 23×23, and 13×13 for the three HSI datasets, respectively.

Then, for the value of K, we study its effects on classification performance in the range of $\{3, 5, 10, 15, 20\}$, and the experimental results are reported in Fig. 5(c). Our APDCLNN model can produce quasi-optimal classification accuracy when K is set to 5 for the three HSI datasets.

To illustrate complementary information and mutual effects between hand-crafted and deep features on HSI classification, an MAMLF module is designed, where the 3-D Gabor filter is used to extract hand-crafted features. According to [4] and [37], to simplify the experimental analysis, the orientations θ and ϕ take the values of $\{0, (\pi/n), (2\pi/n), \ldots, ((n-1)\pi/n)\}$, and $n = 4$. After removing some redundant information, the number (N) of 3-D Gabor filters is 13 for each frequency f. Then, f is searched from $\{0.5, 0.25, 0.125, 0.0625\}$, which is applied as the filterbank to extract hand-crafted spatial–spectral features. The experimental results for analyzing the influences of different frequency f on classification performance are given in Fig. 5(d). From Fig. 5(d), the quasi-optimal value of f is 0.5, 0.5, and 0.25 for the three datasets, respectively.

Furthermore, the effects of different M in the two PDConvLSTM2D layers are analyzed, where M is searched from $\{16, 32\}$, $\{32, 64\}$, and $\{64, 128\}$. From Table I, the optimal combination of M is $\{32, 64\}$ for the three HSI datasets.

TABLE I
SENSITIVITY ANALYSIS UNDER DIFFERENT VALUES OF M

M	Indian Pines	University of Pavia	Houston Data
$\{16, 32\}$	74.76	81.47	79.51
$\{32, 64\}$	**78.86**	**83.85**	**82.19**
$\{64, 128\}$	74.01	81.97	80.17

TABLE II
SENSITIVITY COMPARISON AND ANALYSIS OF DIFFERENT VALUES OF THE DROPOUT OPERATION

Dropout	Indian Pines	University of Pavia	Houston Data
0.4	77.23	80.65	80.26
0.5	**78.86**	**83.85**	**82.19**
0.6	77.31	81.13	81.16

TABLE III
SENSITIVITY ANALYSIS UNDER DIFFERENT VALUES OF λ

λ	0	0.5	1	1.5	2
Indian Pines	74.62	**78.86**	77.93	75.54	76.99
University of Pavia	79.09	**83.85**	80.91	76.53	65.79
Houston Data	77.49	**82.19**	81.36	80.28	80.22

To reduce the occurrence of overfitting, the dropout operation is applied to inactivate part of the input in the GAP layer. Regarding the value of dropout operation, it is selected from a set $\{0.4, 0.5, 0.6\}$. From Table II, for the three HSI datasets, the quasi-optimal value of dropout operation is 0.5.

To adaptively refine the classification results, an attention loss subnetwork is designed, which is combined with the classification loss subnetwork to build a multitask loss function for the optimization of our APDCLNN using λ as balancing parameter in (12). Its optimal value is selected from $\{0, 0.5, 1, 1.5, 2\}$. According to Table III, the quasi-optimal value of λ is 0.5 for the three HSI datasets.

In addition, APDCLNN is trained for 1500 iterations with the learning rate lr in an end-to-end way, and the batch size is set to 32. From Fig. 5(e), the quasi-optimal performance is obtained when lr is set to 0.0005. More detailed parameter settings for the three HSI datasets are shown in Tables IV–VI.

C. Classification Performance

Tables VII–IX show comparison of the classification performance of our APDCLNN with other algorithms. We can find that APDCLNN consistently outperforms all the competitors

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TABLE IV
PARAMETER SETTINGS FOR THE INDIAN PINES DATASET

Layer Name	Kernel Size	Output Size for SSPDCL2DNN	Output Size for GFEB
Input Layer		$23 \times 23 \times 5 \times 1$	
PDConvLSTM2D	4×4	$23 \times 23 \times 5 \times 32$	
3-D Gabor Filters			$23 \times 23 \times 5 \times 13$
MAFB		$23 \times 23 \times 5 \times 64$	
MaxPooling3D	$2 \times 2 \times 2$	$12 \times 12 \times 3 \times 64$	
PDConvLSTM2D	5×5	$12 \times 12 \times 3 \times 64$	
MaxPooling3D	$2 \times 2 \times 2$	$6 \times 6 \times 2 \times 64$	
Dropout Layer	0.5	$6 \times 6 \times 2 \times 64$	
GAP Layer		64	
OSL Layer		16	

TABLE V
PARAMETER SETTINGS FOR THE UNIVERSITY OF PAVIA DATASET

Layer Name	Kernel Size	Output Size for SSPDCL2DNN	Output Size for GFEB
Input Layer		$23 \times 23 \times 5 \times 1$	
PDConvLSTM2D	3×3	$23 \times 23 \times 5 \times 32$	
3-D Gabor Filters			$23 \times 23 \times 5 \times 13$
MAFB		$23 \times 23 \times 5 \times 64$	
MaxPooling3D	$2 \times 2 \times 2$	$12 \times 12 \times 3 \times 64$	
PDConvLSTM2D	4×4	$12 \times 12 \times 3 \times 64$	
MaxPooling3D	$2 \times 2 \times 2$	$6 \times 6 \times 2 \times 64$	
Dropout Layer	0.5	$6 \times 6 \times 2 \times 64$	
GAP Layer		64	
OSL Layer		9	

TABLE VI
PARAMETER SETTINGS FOR THE HOUSTON DATASET

Layer Name	Kernel Size	Output Size for SSPDCL2DNN	Output Size for GFEB
Input Layer		$13 \times 13 \times 5 \times 1$	
PDConvLSTM2D	3×3	$13 \times 13 \times 5 \times 32$	
3-D Gabor Filters			$13 \times 13 \times 5 \times 13$
MAFB		$13 \times 13 \times 5 \times 64$	
MaxPooling3D	$2 \times 2 \times 2$	$7 \times 7 \times 3 \times 64$	
PDConvLSTM2D	5×5	$7 \times 7 \times 3 \times 64$	
MaxPooling3D	$2 \times 2 \times 2$	$4 \times 4 \times 2 \times 64$	
Dropout Layer	0.5	$4 \times 4 \times 2 \times 64$	
GAP Layer		64	
OSL Layer		15	

for the three HSI datasets. Specifically, owing to the ability for solving the problem of processing high-dimensional small-size-sample data and insensitivity to the "curse of dimensionality", SVM-CK, by incorporating a composite kernel into the SVM classifier, yields the second best classification performance by jointly learning the spatial–spectral information, which is usually considered as a strong baseline under the case of small-sample classification [65]. By taking the 3-D cube of HSI data as the input directly, SSCL3DNN [17], 3DG-CNN [37], PCNN [66], and 3DOC-SSAN [67] are superior to some methods of separately learning spatial and spectral information (i.e., SSCL2DNN, SSDCL2DNN, SSPDCL2DNN). With the help of deformable convolution and its CV version, both SSDCL2DNN and SSPDCL2DNN improve the classification performance with stronger feature extraction capabilities than SSCL2DNN. Furthermore, with collaborative learning of hand-crafted and deep features, APDCLNN achieves the best classification accuracy, which is

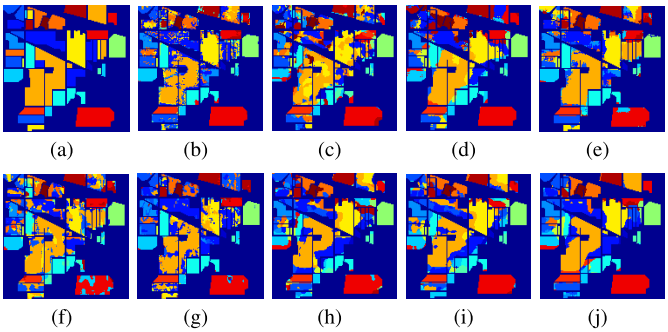

Fig. 6. Classification maps for the Indian Pines dataset. (a) Ground-truth map. (b) SVM-CK. (c) SSCL2DNN. (d) SSCL3DNN. (e) PCNN. (f) 3DOC-SSAN. (g) 3DG-CNN. (h) SSDCL2DNN. (i) SSPDCL2DNN. (j) APDCLNN.

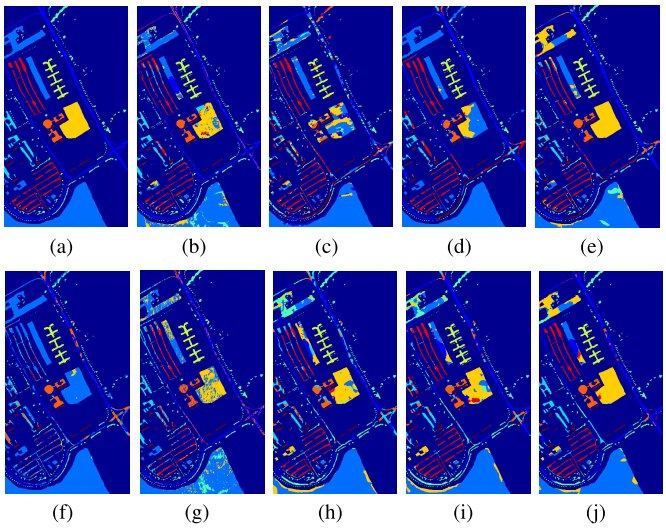

Fig. 7. Classification maps for the University of Pavia data. (a) Ground-truth map. (b) SVM-CK. (c) SSCL2DNN. (d) SSCL3DNN. (e) PCNN. (f) 3DOC-SSAN. (g) 3DG-CNN. (h) SSDCL2DNN. (i) SSPDCL2DNN. (j) APDCLNN.

consistent with the conclusion in [32]–[35]. In addition, 3-D Gabor convolutional neural network (3DG-CNN) uses the 3-D Gabor filter to modulate the random initialization method of convolution kernel, enabling the CNN model to generate spatial–spectral features with better robustness to direction and scale changes [37], which, however, ignores the description ability of hand-crafted features extracted by the 3-D Gabor filter to attribute information of land covers. Different from it, our APDCLNN introduces the idea of manual attention learning to use complementary information and mutual influence between hand-crafted and deep features, improving its training quality and classification performance, especially in the case of small samples. All these demonstrate the feasibility and superiority of APDCLNN, and also illustrate the effectiveness of the MAMLF module on integrating hand-crafted and deep features for HSI classification.

More specifically, to enhance the scale-aware ability of ConvLSTM2D cell, deformable convolution is applied to build a DConvLSTM2D cell, with which an SSDCL2DNN model is proposed to adaptively extract the scale-enhanced spatial–spectral features. Compared with SSCL2DNN [17], it can obtain 3.03%, 7.28%, and 4.72% gains in OA for the three HSI datasets, respectively. Furthermore, considering that

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TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE INDIAN PINES DATASET

Class	SVM-CK	SSCL2DNN	SSCL3DNN	PCNN	3DOC-SSAN	3DG-CNN	SSDCL2DNN	SSPDCL2DNN	APDCLNN
1	97.78	98.15	**100.00**	99.72	90.56	98.06	97.78	97.50	99.72
2	59.56	38.65	53.30	60.28	51.43	**69.68**	51.30	53.19	68.43
3	64.95	46.34	50.37	49.68	49.89	55.98	51.73	54.12	**66.96**
4	80.44	79.88	88.63	94.32	83.44	84.41	87.44	91.15	**95.29**
5	87.53	61.95	76.58	84.08	64.00	**90.61**	65.03	65.43	73.68
6	94.39	65.56	90.17	**97.10**	80.25	92.26	66.03	66.42	83.58
7	98.89	**100.00**	**100.00**	**100.00**	**100.00**	99.44	**100.00**	**100.00**	**100.00**
8	93.59	94.59	**100.00**	99.25	96.90	95.64	97.86	98.55	99.98
9	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	92.00	96.00	**100.00**
10	74.07	66.22	75.36	69.10	62.52	67.06	64.91	64.61	**75.97**
11	59.56	50.46	59.66	69.73	55.68	59.21	52.13	60.33	**73.49**
12	57.63	60.83	64.97	58.25	48.44	71.34	60.50	60.14	**75.15**
13	97.74	96.58	**99.79**	99.38	93.74	98.31	92.97	97.13	96.56
14	**92.10**	73.86	88.29	86.78	78.58	89.10	75.02	76.48	84.38
15	78.62	84.13	87.66	89.60	78.01	82.39	82.55	84.60	**90.03**
16	98.31	91.16	**100.00**	99.88	99.16	99.40	96.87	95.90	98.67
OA	73.18	60.59	71.28	74.47	64.55	73.87	63.62	66.51	**77.82**
AA	83.45	75.52	83.42	84.82	77.04	84.60	77.13	78.85	**86.37**
κ	69.79	56.12	67.83	71.11	60.16	71.08	59.39	62.45	**75.02**

TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE UNIVERSITY OF PAVIA DATASET

Class	SVM-CK	SSCL2DNN	SSCL3DNN	PCNN	3DOC-SSAN	3DG-CNN	SSDCL2DNN	SSPDCL2DNN	APDCLNN
1	73.74	49.43	69.89	72.00	68.76	70.66	64.69	69.39	**82.05**
2	**83.09**	76.89	82.44	79.77	75.89	77.71	67.03	68.97	76.51
3	74.47	39.38	65.10	71.32	68.88	67.20	59.08	67.64	**84.19**
4	89.04	62.48	90.48	**96.82**	90.44	89.31	81.01	79.88	82.59
5	**99.97**	84.42	94.83	98.85	99.04	99.85	96.96	97.84	97.89
6	78.99	41.29	82.37	**91.66**	80.60	80.06	78.31	78.29	90.39
7	88.21	61.16	89.32	82.89	80.61	83.05	83.43	86.73	**96.45**
8	77.39	66.68	84.71	73.09	69.69	71.49	84.54	88.42	**94.09**
9	81.88	83.46	86.91	96.10	87.85	**98.99**	81.35	77.13	90.88
OA	81.33	64.59	81.11	81.25	76.64	78.00	71.87	74.15	**82.93**
AA	82.97	62.80	82.90	84.72	81.20	82.04	77.38	79.36	**88.34**
κ	75.87	54.45	75.65	76.04	70.23	71.88	64.77	67.56	**78.31**

TABLE IX
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE HOUSTON DATASET

Class	SVM-CK	SSCL2DNN	SSCL3DNN	PCNN	3DOC-SSAN	3DG-CNN	SSDCL2DNN	SSPDCL2DNN	APDCLNN
1	**90.81**	87.51	85.98	85.18	91.03	88.76	87.45	87.32	90.07
2	88.04	62.32	81.27	78.78	83.49	86.75	72.77	82.14	**88.70**
3	88.96	85.34	95.68	93.14	93.82	**98.20**	93.08	93.07	92.99
4	86.97	68.18	84.80	81.60	83.70	88.44	80.43	81.84	**89.74**
5	99.12	84.50	95.05	98.26	94.34	**99.86**	89.26	89.75	94.87
6	80.38	59.94	73.38	84.30	80.48	73.41	65.70	69.49	**90.96**
7	65.84	56.63	73.86	75.37	71.17	69.37	72.42	73.29	**83.39**
8	53.99	43.52	**60.19**	52.70	64.42	52.72	46.43	48.92	58.02
9	76.99	55.06	70.21	67.48	66.40	63.82	54.66	58.41	**78.16**
10	76.95	60.25	68.82	**77.03**	73.02	73.06	57.67	61.16	71.25
11	69.00	59.92	61.57	68.79	**70.79**	69.67	55.76	56.66	67.36
12	72.27	47.68	61.91	**79.17**	78.57	73.47	53.94	60.04	68.42
13	72.79	56.86	57.60	**90.11**	87.21	76.80	69.37	77.78	86.14
14	91.99	90.77	**98.85**	97.92	97.13	97.18	90.50	95.84	96.32
15	**95.94**	70.09	84.48	94.89	89.75	98.61	71.81	71.90	94.41
OA	79.57	64.50	75.97	79.20	79.86	79.07	69.22	72.10	**81.31**
AA	80.67	65.90	76.91	81.65	81.69	80.68	70.75	73.84	**83.39**
κ	77.92	61.70	74.04	77.53	78.25	77.38	66.79	69.90	**79.83**

CV-DNNs have the advantages of easier optimization, better generalization characteristic, and faster learning than their RV versions [38]–[51], the DConvLSTM2D cell is extended to the PDConvLSTM2D cell using the complex number operation to improve its ability of modeling the spectral correlation, and an SSPDCL2DNN model can be obtained to jointly learn the scale- and spectral-enhanced spatial–spectral features for HSI classification. Thus, SSPDCL2DNN further improves the classification accuracy of SSDCL2DNN by 2.89%, 2.28%, and 2.88% in OA for the three HSI datasets, respectively.

Based on the enhanced spatial–spectral features from SSPDCL2DNN, by integrating hand-crafted features from

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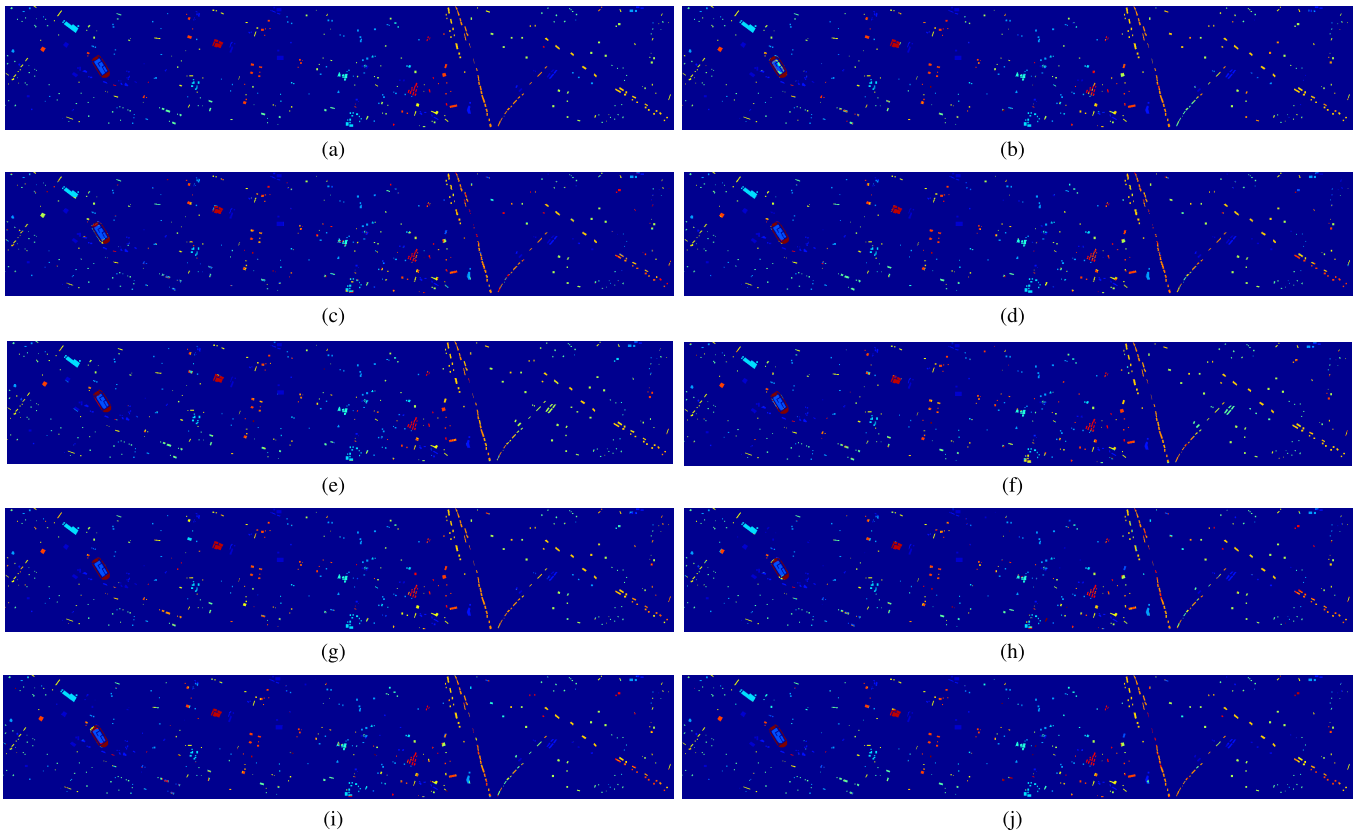

Fig. 8. Classification maps for the Houston dataset. (a) Ground-truth map. (b) SVM-CK. (c) SSCL2DNN. (d) SSCL3DNN. (e) PCNN. (f) 3DOC-SSAN. (g) 3DG-CNN. (h) SSDCL2DNN. (i) SSPDCL2DNN. (j) APDCLNN.

the 3-D Gabor filters, our APDCLNN achieves 11.31%, 8.78, and 9.21% gains in OA upon SSPDCL2DNN, 6.54%, 1.82, and 5.34% improvements over SSCL3DNN, while 3.35%, 1.68%, and 1.59% gains than PCNN for the three datasets, respectively. Compared with SVM-CK, APDCLNN can obtain 4.64%, 1.60%, and 1.74% gains for the three HSI datasets, respectively. In particular, the improvements in OA achieved by APDCLNN are 3.95%, 4.93%, and 2.24% for the three HSI datasets, respectively, when compared with 3DG-CNN. These suggest that exploring the complementary information between hand-crafted and deep features can be effective to improve the network optimization and classification performance of DNNs, especially in the case of small samples for HSIs. Particularly, after same data preprocessing, 3DOC-SSAN [67] fails to generate the expected results due to the insufficient use of spatial–spectral information, especially under the small-sample classification of HSIs.

From the perspective of visual analysis for the three HSI datasets, Figs. 6–8 visualize the classification maps of all considered algorithms corresponding to Tables VII–IX. As shown in Figs. 6–8, the classification maps yielded by our APDCLNN are the closest to the ground-truth maps, especially for class 2, class 10, class 11, and class 12 in Fig. 6, class 1, class 3, and class 8 in Fig. 7, and class 6 and class 7 in 8, which is consistent with the conclusions in Tables VI–VIII.

To evaluate the impact of the training size, 20, 30, and 40 samples of each class are randomly selected from the three HSI datasets for training. Particularly, for class 7 and class 9 in the Indian Pines dataset, their sample numbers are fixed to 10. The OA curves of all the methods are reported in Fig. 9. We can observe that our APDCLNN can generate the best classification accuracy on all the cases for the three HSI datasets.

Moreover, to further illustrate the classification performance over background pixels for all the considered methods, by considering the characteristics of scene complexity and region heterogeneity, Figs. 10 and 11 list their full scene classification maps under 10 and 40 samples of each class for the Houston dataset, respectively. From Fig. 10, compared with other comparison methods, APDCLNN can learn more discriminative spatial–spectral features by exploring the complementary information between different modality features, providing relatively better full classification maps. Furthermore, with the increase in the training sample number, the quality of classification maps of all the considered methods is significantly improved, where more training samples increase their fitting ability and classification performance for the background pixels and the borders of different regions. Particularly, the full classification map of APDCLNN contains clearer boundaries and less noise. The experimental results demonstrate the effectiveness of the proposed APDCLNN model for HSI classification.

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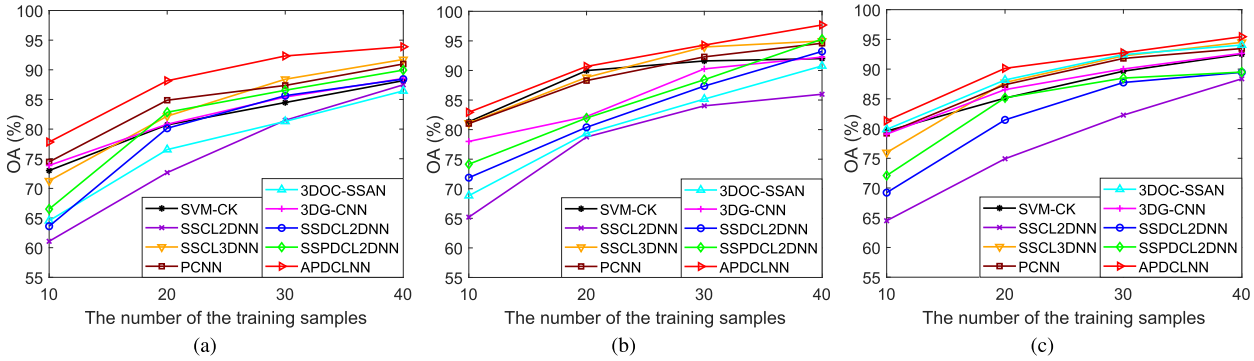

Fig. 9. OA (%) of all considered algorithms under different numbers of training samples. (a) Indian Pines. (b) University of Pavia. (c) Houston Data.

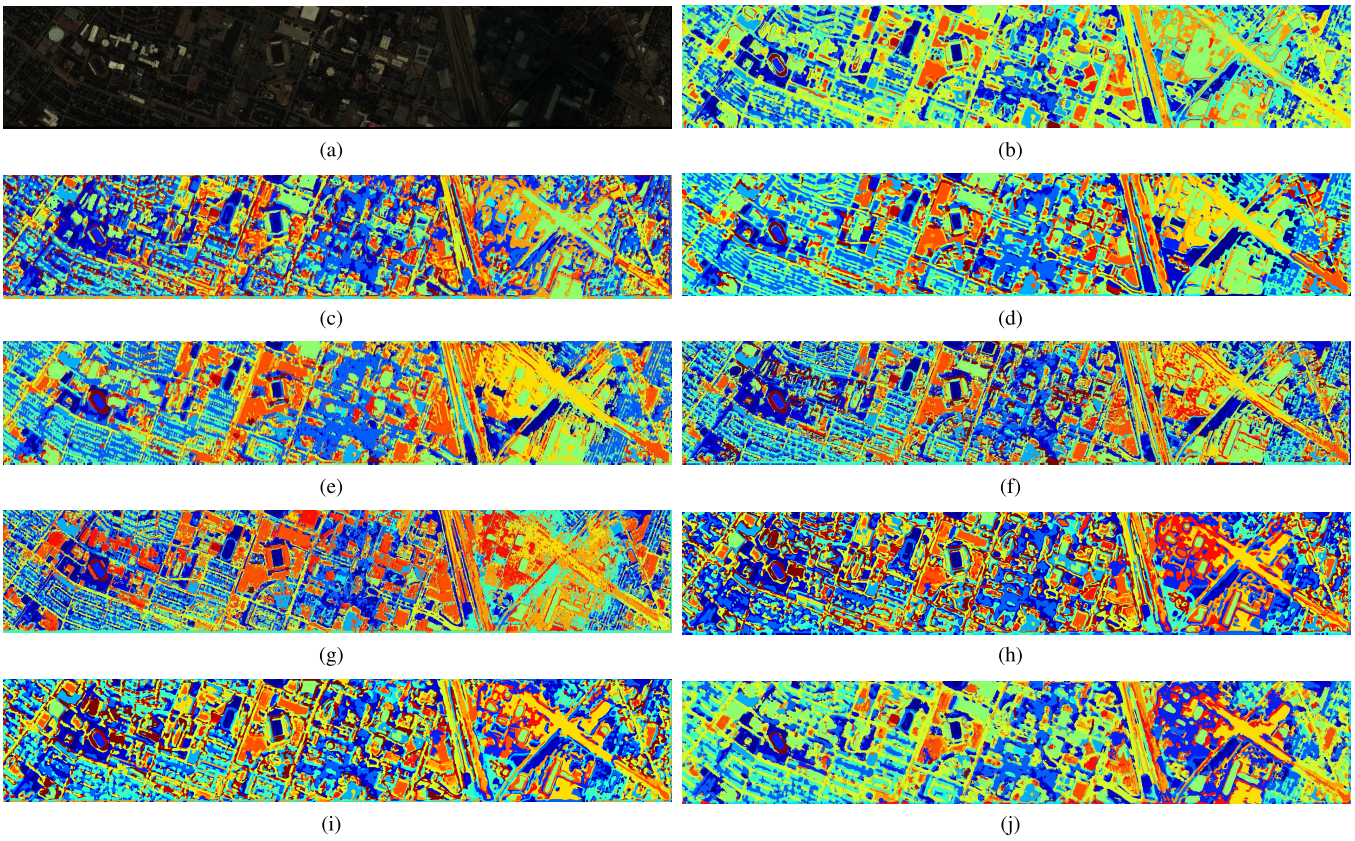

Fig. 10. Full classification maps for the Houston dataset (10 samples of each class). (a) False-color map. (b) SVM-CK. (c) SSCL2DNN. (d) SSCL3DNN. (e) PCNN. (f) 3DOC-SSAN. (g) 3DG-CNN. (h) SSDCL2DNN. (i) SSPDCL2DNN. (j) APDCLNN.

D. Ablation Study

To illustrate the effectiveness of the PDConvLSTM2D cell, the MAMLF module, and the attention loss subnetwork, we perform a detailed ablation study to measure their contributions to the classification accuracy of APDCLNN. In particular, the proposed APDCLNN with and without each component are represented as proposed (with) and proposed (without) for convenience. SSCL2DNN [17] is considered as the baseline.

1) Analysis of the PDConvLSTM2D Cell: To enhance the feature extraction ability of ConvLSTM2D for scale information and correlation information of HSI data, a PDConvLSTM2D cell is developed using deformable convolution in the CV domain. In particular, to analyze its contribution for HSI classification, the proposed APDCLNN is compared with its variants [i.e., the proposed (without PDConvLSTM2D)] where the second PDConvLSTM2D layer is replaced with the CV-ConvLSTM2D layer that is directly improved by the CV convolutional layer [48]. From Table X, compared with the variant, the PDConvLSTM2D cell can bring 3.29%, 1.47%, and 3.65% improvements in OA to our APDCLNN, respectively. Specifically, the experimental results in the last three lines of Table X illustrate that the PDConvLSTM2D cell and the MAMLF module can contribute to the classification performance of APDCLNN for HSIs.

2) Architecture Analysis of the MAMLF Module: By jointly considering the contributions of hand-crafted and deep

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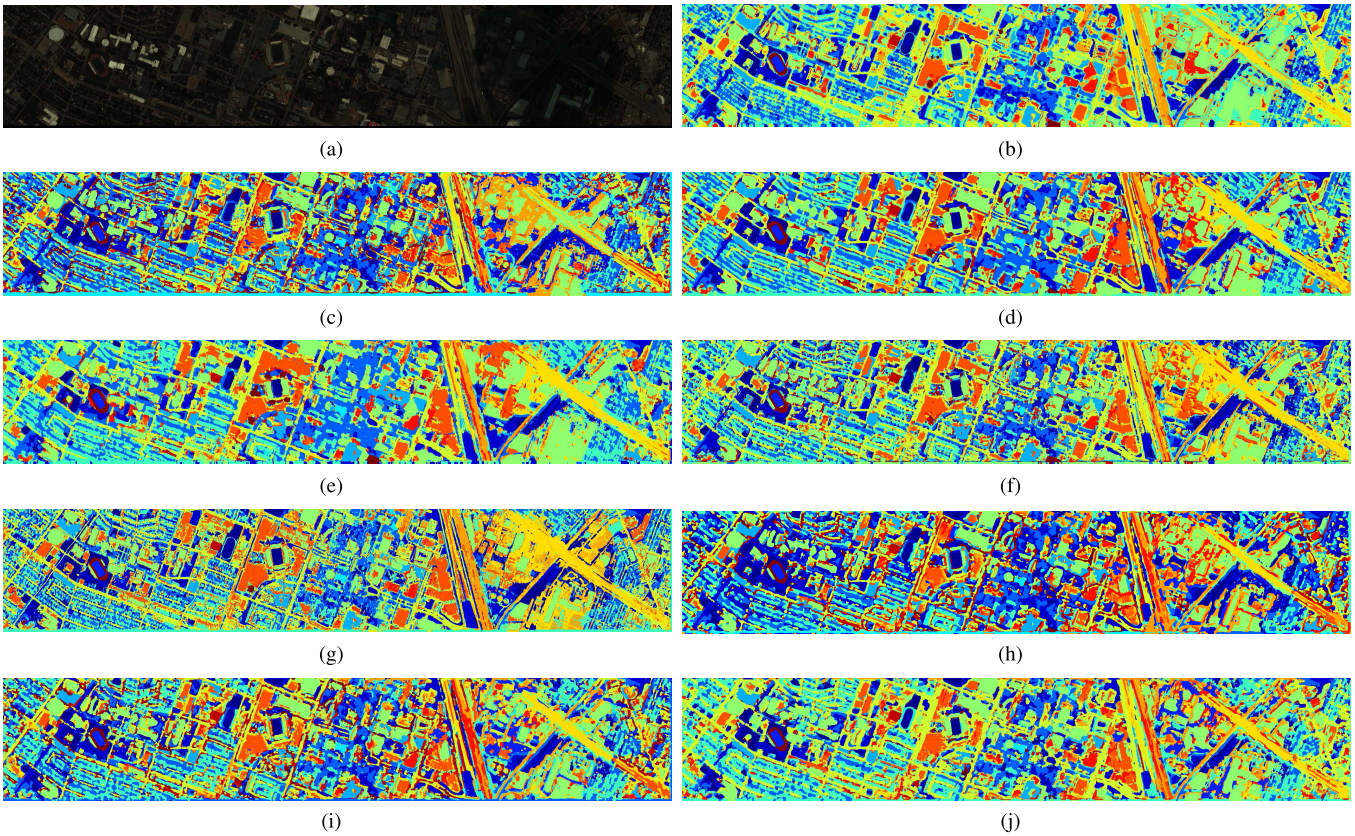

Fig. 11. Full classification maps for the Houston dataset (40 samples of each class). (a) False-color map. (b) SVM-CK. (c) SSCL2DNN. (d) SSCL3DNN. (e) PCNN. (f) 3DOC-SSAN. (g) 3DG-CNN. (h) SSDCL2DNN. (i) SSPDCL2DNN. (j) APDCLNN.

TABLE X
PERFORMANCE ANALYSIS OF THE PDCONVLSTM2D CELL

Models	Indian Pines		University of Pavia		Houston Data	
	OA	κ	OA	κ	OA	κ
baseline	60.59	56.12	64.59	54.45	64.50	61.70
SSDCL2DNN	63.62	59.39	71.87	64.77	69.22	66.79
SSPDCL2DNN	66.51	62.45	74.15	67.56	72.10	69.90
proposed(without)	74.53	71.33	81.46	76.73	77.66	75.88
proposed(with)	**77.82**	**75.02**	**82.93**	**78.31**	**81.31**	**79.83**

TABLE XI
STRUCTURE ANALYSIS OF THE MAMLF MODULE

Models	Indian Pines		University of Pavia		Houston Data	
	OA	κ	OA	κ	OA	κ
baseline	60.59	56.12	64.59	54.45	64.50	61.70
proposed(without)	66.51	62.45	74.15	67.56	72.10	69.90
proposed(with)	**77.82**	**75.02**	**82.93**	**78.31**	**81.31**	**79.83**

TABLE XII
ARCHITECTURE STUDY OF THE MAFB SUBMODULE

Models	Indian Pines		University of Pavia		Houston Data	
	OA	κ	OA	κ	OA	κ
baseline	60.59	56.12	64.59	54.45	64.50	61.70
proposed(without)	70.23	66.57	76.59	70.50	74.92	72.91
proposed(with)	**77.82**	**75.02**	**82.93**	**78.31**	**81.31**	**79.83**

spatial–spectral features to HSI classification, a learnable MAMLF module is built. Some experiments are put into effect to evaluate its superiority on accuracy by adding and removing it from our APDCLNN. Actually, the proposed (without MAMLF) model in Table XI is SSPDCL2DNN. From Table XI, the MAMLF module can bring at least 8.50% gains to the whole model, which indicates that it can effectively integrate hand-crafted features with interpretable spatial–spectral information into the training process of our APDCLNN and reduce the impact of overfitting from limited samples to a certain extent, thus further improving its classification performance, which is consistent with the results in Table X.

3) Structure Study of the MAFB Submodule: Hand-crafted features can provide deep features with some detailed and interpretable information [32]–[35]. However, directly cascading these two modality features may not be able to effectively exchange their complementary information and mutual effects. As such, a ConvLSTM3D-based MAFB submodule is designed to fuse them. We compare the proposed APDCLNN with its variant [i.e., proposed (without MAFB)] where MAFB is replaced with the concatenation operation, and the corresponding experimental results are reported in Table XII. Compared with this variant, APDCLNN can obtain at least 6.00% gains in OA for the three HSI datasets, thus showing the superiority of the MAFB submodule over direct cascade operation in integrating and capturing the complementarity of different modality features to improve the classification performance of APDCLNN for HSIs.

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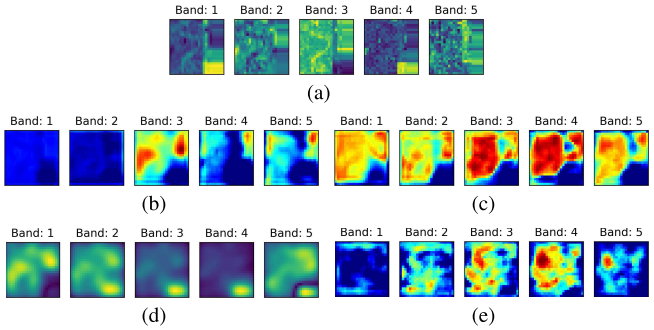

Fig. 12. Visualization results of attention masks in the MAFB module for the Indian Pines dataset. (a) Input data. (b) Deep feature without attention. (c) Deep feature with attention. (d) Hand-crafted feature without attention. (e) Hand-crafted feature with attention.

4) Effectiveness of the Attention Loss Subnetwork: To refine the final classification results adaptively, an attention loss subnetwork is designed to integrate with the classification loss subnetwork, thus forming a multitask loss function for training and optimization of APDCLNN in an end-to-end way. Actually, from (12), the attention loss subnetwork is controlled by the parameter λ. When λ equals 0, it means that the attention loss subnetwork is not considered. From Table III, compared with the variant with the attention loss subnetwork removed, our attention loss subnetwork can bring at least 4.00% gains to APDCLNN. More detailed experimental results are reported in Table III, which illustrate the effectiveness of our attention loss subnetwork in HSI classification.

5) Visualizations for the Attention Masks: According to Section III-C, for deep and hand-crafted features, the corresponding attention masks in the MAFB submodule present the fourth-order tensor with a size of $K \times w \times h \times C'$. It is very difficult to draw their visualizations directly. Therefore, to visualize the working mechanism of the above attention masks, we extract a channel map from two inputs without attention [i.e., $F^{(1)}$ and $F^{(2)}$ in (8)] and the corresponding outputs with attention [i.e., $\alpha^{(2)} \otimes F^{(1)'}$ and $\alpha^{(1)} \otimes F^{(2)'}$ in (9)] from the MAFB submodule, respectively, obtaining four third-order tensors with a size of $K \times w \times h$. After that, we show some spatial attention maps along with different spectral dimensions to see whether they highlight clear semantic areas.

Based on the above analysis, visualizations of attention masks in the MAFB submodule are shown in Fig. 12 for the Indian Pines dataset. Fig. 12(a) illustrates an example of input data X^H, and Fig. 12(b) and (d) are the deep and hand-crafted features, respectively, whose corresponding outputs with attention are given in Fig. 12(c) and (e). Compared with two modality features without attention in Fig. 12 (b) and (d), we can observe that after imposing mutual attention learning, both the modality features in Fig. 12 (c) and (e) can capture relatively clear semantic similarity of each class in the spectral dimension, while the response of specific semantics is noticeable and some interference information can be suppressed in spatial dimensions. Therefore, with the designed MAFB submodule, the proposed APDCLNN model can effectively highlight the response and detailed information of specific semantic area, thus improving its spatial–spectral feature extraction and classification performance for HSIs.

TABLE XIII

ANALYSIS FOR THE MODEL COMPLEXITY AND THE TIME COMPLEXITY OVER DIFFERENT DEEP MODELS

	Indian Pines		University of Pavia		Houston Data	
	Size	Time	Size	Time	Size	Time
baseline	9.63	3.48	9.79	23.93	6.54	3.98
SSCL3DNN	37.30	11.89	37.30	52.13	29.60	5.30
PCNN	7.29	0.77	7.27	2.09	4.86	1.07
3DOC-SSAN	13.70	1.05	13.50	3.65	13.70	1.57
3DG-CNN	20.70	9.76	20.70	36.93	20.70	14.08
SSDCL2DNN	10.90	37.27	10.00	153.50	10.80	19.92
SSPDCL2DNN	21.60	71.28	19.90	293.59	21.40	36.19
APDCLNN	37.70	96.07	35.40	412.70	33.60	47.74

6) Analysis for the Model Complexity and the Time Complexity: Based on the parameter settings in Section IV-B, some experiments for analyzing the model complexity and the time complexity of different deep models have been explored on the three HSI datasets, whose results are reported in Table XIII. In particular, "Size" refers to the model storage size (MB), and "Time" means the testing time (s), which is the average running time of ten times.

From Table XIII, compared with the convolutional-based PCNN, 3DOC-SSAN, and 3DG-CNN, the ConvLSTM2D- or ConvLSTM3D-based models (i.e., baseline, SSCL3DNN) have higher model storage size and time complexity due to the design of multiple gate structures for modeling spectral correlation. Although SSCL3DNN, SSDCL2DNN, and SSPDCL2DNN achieve better classification accuracy for HSIs than baseline, they have more underlying parameters and higher computational complexity with the implementation of additional 3-D convolution, deformable convolution, and CV deformable convolution operations to extract more expressive spatial–spectral features for HSI classification, respectively.

Based on SSPDCL2DNN, the MAFB submodule is designed to fuse the multimodality features, leading to more network parameters for the proposed APDCLNN. Furthermore, the introduction of hand-crafted features provides an attempt to solve the problems of poor interpretability of deep features and high optimization difficulty of DNNs for HSI classification to a certain degree; however, the additional processes for hand-crafted feature extraction and multimodality feature fusion further increase the computational complexity of APDCLNN. As such, the parameter and time complexities of the whole model are the highest. The lightweight design (even lossless compression) of complex number operation and fusion module, and the simplification of hand-crafted feature extraction and fusion processes are our future research directions to overcome the problem of high complexity for the proposed APDCLNN, so as to be better used in practical applications.

V. CONCLUSION

In this article, a novel APDCLNN model has been proposed for HSI classification, providing an effective way to integrate the traditional hand-crafted features into deep semantic

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features for improving the classification performance, especially in the case of small samples. First, to enhance the feature extraction abilities for scale information and spectral correlation of HSIs, by combining the respective advantages of deformable convolution and complex number operations, a new PDConvLSTM2D cell is designed, with which an SSPDCL2DNN model is built. Then, to effectively supervise the training process and reduce the impact of overfitting from limited samples of SSPDCL2DNN, a ConvLSTM3D-based MAMLF module is further developed to adaptively exchange the complementarity and mutual effects of hand-crafted features and deep features, thus extracting more discriminative spatial–spectral features for HSI classification. Finally, an attention loss subnetwork is collaborated with the classification loss subnetwork to construct a multitask loss function for adaptively optimizing the classification performance of APDCLNN. We evaluate our APDCLNN on three HSI datasets and consistently improve the classification accuracy in comparison to other deep learning methods.

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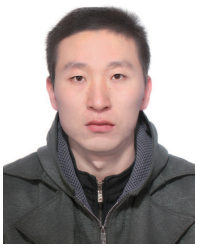

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